Data-Driven Design & Analysis of Structures & Materials (3dasm)

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Homework 8

Deliver a **ZIP folder with two files (PDF and Jupyter notebook)** containing the answers to the questions of this assignment. **UPLOAD to CANVAS in the Assignments section (Homework 8) by the due date.**

Due date: until 11:59pm of day announced in course home page.

This Homework aims to help you reason about logistic regression: why the sigmoid, what the decision boundary of the predicted class is, why the maximum likelihood estimate does not exist when the classes are separable, and what the hyperparameter `C` of scikit-learn means. It is a short homework, so that you have time for the Final Project. It contains questions to be solved with **pen and paper** as well as with **code**: the notebook you submit contains your prompt, the code the agent wrote, its results and your answers.

How to work. Start from the notebook `Assignments/3dasm_Homework8.ipynb`: copy it to `Assignments/your_Assignments/HW8/your_Homework8.ipynb` and work there. As in the lectures, an AI coding agent writes and runs the code, and the notebook is the source of truth: you write the agent's instructions file (`CLAUDE.md`) in a notebook cell, each prompt you give the agent is a notebook cell, and the cells that we provide load each agent's `results.json` from its task folder.

- **Model.** Use Sonnet or Haiku (`set_model("sonnet")` or `set_model("haiku")`); the course's helper does not accept other models.
- **Grading.** The TA runs your notebook **from scratch** (every agent runs again), with the model you declare in it. The regenerated numbers must be the same as the ones you submitted: if they are not, your instructions left something for the agent to guess. Your numbers are also compared with the solution: the homework fixes the dataset, the split and the models, so a precise notebook reproduces them exactly. Your predictions, checks and interpretations are graded as usual; base them on the numbers and figures your notebook stores, not on the agent's summary.
- **Prompts** are written in words. Short expressions that pin the data, the randomness or the method are fine (e.g. `random_state=123`); code that implements the task is not.
- **Your prompts and your instructions file are graded.** In the exercises with the agent, a good prompt defines the problem **uniquely**: the data, the randomness (seeds, and the order of the draws), the model and the method, and what to store and plot. A prompt that leaves choices open makes the outcome depend on what the agent happens to guess, which can change from one run to the next. The TA re-runs your notebook with fresh agents: a number that changes counts as an error, even if your answer was right. The TA also reads your prompts: an ambiguity that happened to reproduce counts as a minor error.
- **Write your prompts in your own words.** A prompt that only points to a question (e.g. "solve exercise 2.2") or that pastes it is not a prompt: that item counts as not answered. You may copy the specifications (numbers, seeds, names of functions).
- Each exercise that says **(pen and paper)** should not be resolved with an AI agent: it trains the reasoning that you need to check what an agent does
- Each exercise that says **(with the agent)** is appropriate to resolve with an agent as we introduced in the lectures.
  - Note: **writing your own code instead of using the agent is allowed** for the exercises signalled as **(with the agent)**. Put your code in the same notebook cell (not in a separate file).
- **Before you submit**, delete the task folders and run the notebook from the start, so that its outputs come from one clean run (this is also what the TA will do when grading).

- **Submit one ZIP file** with your notebook (`.ipynb`), which the TA re-runs, and a PDF of that same notebook after your clean run. If you write the pen and paper answers by hand, add a scan of them as a third file. Do not include the task folders (they are not needed: the notebook is the source of truth).

Exercise 1 (pen and paper)

Binary logistic regression models a class $y \in \{0, 1\}$ with the observation distribution $p(y|\mathbf{x}, \mathbf{w}) = \text{Ber}(y|\rho)$, $\rho = \sigma(a)$, $a = \mathbf{w}^T\boldsymbol{\phi}(\mathbf{x})$, where $\sigma(a) = 1/(1 + e^{-a})$ is the sigmoid.

1. Logistic regression, by hand.

   1.1. **The sigmoid**: show that $\sigma(-a) = 1 - \sigma(a)$ and that $\frac{d\sigma}{da} = \sigma(a)\,(1 - \sigma(a))$. Linear regression used the linear function $\mathbf{w}^T\boldsymbol{\phi}(\mathbf{x})$ directly as the mean of a Gaussian: why can it not be used directly as the parameter ρ of the Bernoulli distribution?

   1.2. **The decision boundary**: let $\boldsymbol{\phi}(\mathbf{x}) = [1, x_1, x_2]^T$. Show that the predicted class (the mode of the PPD) is $\hat{y}^* = 1$ if and only if $w_0 + w_1x_1^* + w_2x_2^* > 0$, so that the decision boundary is a straight line. On which line is $\rho = 0.9$, and on which is $\rho = 0.1$? Show that the distance between these two lines is $2\ln 9/\sqrt{w_1^2 + w_2^2}$, and say what happens to it when all the weights are multiplied by 10.

   1.3. **Separable classes**: suppose that some weights $\mathbf{w}$ classify every training point correctly, with $\mathbf{w}^T\boldsymbol{\phi}(\mathbf{x}_n) > 0$ when $y_n = 1$ and < 0 when $y_n = 0$. Show that the negative log likelihood of $t\mathbf{w}$ decreases as $t > 1$ grows, and tends to 0. What does this say about the maximum likelihood estimate of $\mathbf{w}$ for separable classes? Using 1.2., what does the PPD look like for very large t?

Exercise 2 (with the agent)

In this exercise you measure what the hyperparameter `C` of scikit-learn's `LogisticRegression` does, on two problems built from the Iris dataset (`data_iris.csv`, in the folder `Assignments/data`), each one with two inputs and the bias:

- **Problem A**: all 150 flowers; inputs petal length and petal width; class 1 is *Iris-setosa*, class 0 the other two species.
- **Problem B**: only the 100 flowers of *Iris-versicolor* and *Iris-virginica*; inputs sepal length and sepal width; class 1 is *Iris-virginica*.

Use all the flowers of each problem for training (there is no testing set: this exercise is about the fit, not about the predictions), the raw inputs (no scaling), the `lbfgs` solver with `tol=1e-10` and `max_iter=100000`, and `C` $\in \{10^{-2}, 1, 10^2, 10^4, 10^6\}$.

2. What `C` is.

   2.1. **Before running anything**, look at the flowers of each problem (the notebook plots them for you) and predict: in which problem will the weights (w_1, w_2) keep growing as `C` grows, and why? (You will not lose points for wrong predictions. You will lose points for not writing them.)

   2.2. Write a prompt that fits the ten models and stores, for each problem and each `C`: the weights w_1, w_2, the bias w_0, $\|(w_1, w_2)\|$, the negative log likelihood of the training data (the sum over the flowers, with the natural logarithm), and the training accuracy. Show, for each problem, the flowers with the decision boundary and the lines $\rho = 0.1$ and $\rho = 0.9$ for `C`$= 1$ and `C`$= 10^6$.

   2.3. **Check** (your own code, in a notebook cell, using only the stored weights and the data): compute the gradient of the negative log likelihood, $\boldsymbol{\Phi}^T(\boldsymbol{\rho} - \mathbf{y})$, at each of the ten fits. It is not zero: compare it, component by component, with $-\mathbf{w}/$`C`, where the bias component of $\mathbf{w}$ is set to zero (differences of about 10^{-6} or smaller are the precision of the solver, not a disagreement). Which objective does scikit-learn minimize? Write it, and identify `C` in terms of a Gaussian prior on the weights. Why is the bias treated differently?

2.4. **What can you conclude?** Compare with your predictions of 2.1. and with Exercise 1 (1.2. and 1.3.): why do the weights converge in one problem and not in the other, and how does the width of the band between $\rho = 0.1$ and $\rho = 0.9$ change? In Problem A, compute from the stored weights the probability that a flower with petal length 2.8 cm and petal width 0.9 cm is setosa, for `C`$= 1$ and `C`$= 10^6$. This flower lies in the gap between setosa and the other species: would you trust either probability? What does the prior buy you, and what would you need to choose `C`?